

Project Initialization and Planning Phase

Date	23 June 2026
Team ID	xxxxxx
Project Name	OptiCrop – AI-Based Crop Recommendation System
Maximum Marks	3 Marks

Define Problem Statements (Customer Problem Statement):

Farmers often face challenges in selecting the most suitable crop because they may not have accurate knowledge of soil nutrients and environmental conditions. Choosing an unsuitable crop can result in reduced productivity, financial losses, and inefficient use of agricultural resources. The OptiCrop – AI-Based Crop Recommendation System uses Machine Learning to analyze soil parameters such as Nitrogen (N), Phosphorus (P), Potassium (K), temperature, humidity, pH, and rainfall to recommend the most suitable crop. This enables farmers to make informed decisions.

Example:



Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A farmer with agricultural land.	Select the most suitable crop for cultivation.	I do not know which crop is best for my soil and weather conditions.	Soil nutrients and climate vary from region to region.	Confused about making the right farming decision and worried about crop failure.
PS-2	A new farmer with limited agricultural knowledge.	Improve crop yield and maximize profit.	I lack in analyzing soil and environmental conditions	Professional agricultural guidance is not always easily	Uncertain about crop selection and concerned about financial losses.